

- a. Name of two GenAI programs: CoPilot and Claude
- b. How you accessed them (e.g., Google query, CoPilot on my computer): directly through their official websites on new accounts
- c. Prompt used. (You must use the same prompt for both GenAIs.):

## Objective

Write a C program that picks a secret number between 1 and 10.

The user gets up to 3 tries to guess the number.

## Requirements

- Use a fixed secret number (for example, 7).
- Use printf to prompt the user and scanf to read their guess.
- Use a while or for loop to let the user guess up to 3 times.
- Use if-else statements to check if the guess is too high, too low, or correct.
- Stop the loop immediately if the user guesses the correct number.
- Print a final win or lose message.

## Example Output

text

Guess a number between 1 and 10.

Attempt 1/3. Enter your guess: 5

Too low! Try again.

Attempt 2/3. Enter your guess: 8

Too high! Try again.

Attempt 3/3. Enter your guess: 7

Correct! You win!

d. Generated code from both GenAIs.

CoPilot:

```
#include <stdio.h>
```

```
int main() {
```

```
    int secret = 7;    // fixed secret number
```

```
    int guess;
```

```
    int attempts = 3;
```

```
    printf("Guess a number between 1 and 10.\n");
```

```
    for (int i = 1; i <= attempts; i++) {
```

```
        printf("Attempt %d/%3d. Enter your guess: ", i, attempts);
```

```
        scanf("%d", &guess);
```

```
        if (guess == secret) {
```

```
            printf("Correct! You win!\n");
```

```
            return 0; // stop immediately
```

```
        } else if (guess < secret) {
```

```
            printf("Too low! Try again.\n");
```

```
        } else {
```

```
            printf("Too high! Try again.\n");
```

```
        }
```

```
    }
```

```
    // If loop finishes without a correct guess:
```

```
printf("Sorry, you lose! The correct number was %d.\n", secret);
```

```
return 0;
```

```
}
```

Claude:

```
#include <stdio.h>
```

```
int main(void) {
```

```
    int secret = 7;
```

```
    int guess;
```

```
    int won = 0;
```

```
    printf("Guess a number between 1 and 10.\n");
```

```
    for (int attempt = 1; attempt <= 3; attempt++) {
```

```
        printf("Attempt %d/3. Enter your guess: ", attempt);
```

```
        scanf("%d", &guess);
```

```
        if (guess == secret) {
```

```
            printf("Correct! You win!\n");
```

```
            won = 1;
```

```
            break;
```

```
        } else if (guess < secret) {
```

```
            printf("Too low! Try again.\n");
```

```
        } else {
```

```
            printf("Too high! Try again.\n");
```

```

    }
}

if (!won) {
    printf("Out of tries! The number was %d. You lose.\n", secret);
}

return 0;
}

```

e. Analyze the code in terms of correctness, execution time, space complexity, and maintainability as explained in the “GenAI Code Comparison” lecture.

- a. Correctness: Both of them only let you make 2 guesses because the bounds on the for loop are wrong. They both treat letters like numbers. Letters result in the “too low” message, while multiple letters result in a “too high” message.
- b. Execution time: They are both  $O(n)$  because of the for loop.
- c. Space complexity: The only difference in how they use variables is that Claude has a variable to store whether or not you won, and CoPilot has a separate variable to store the number of attempts that are allowed
- d. Maintainability: CoPilot has three comments, and Claude has none. Neither of them have enough comments for a future programmer to be able to easily maintain this.

f. Select one to use as the basis for this assignment and justify why you picked it.

The code is pretty much the same on both. I decided to use CoPilot’s code because the formatting on the prints is a little nicer.

g. Explain what you did to improve it in terms of correctness, execution time,

space complexity, and maintainability.

I removed the win variable because it was unnecessary. I changed the break after a win to a return so that if the for loop ended, it would automatically mean that you lost the game. I also added comments explaining each line.